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1 Likelihoods

What is a Likelihood?

$$y_i \stackrel{\text{iid}}{\sim} N(x_i^\top \beta, \sigma^2)$$

$$f(y_i) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2\sigma^2}(y_i - x_i^\top \beta)^2}$$

That's just a height given a certain y_i . The likelihood is a product of function heights.

$$f(y_1 \cap y_2) = f(y_1)f(y_2) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2\sigma^2}(y_1 - x_1^\top \beta)^2} \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2\sigma^2}(y_2 - x_2^\top \beta)^2}$$

Likelihoods often mentioned as probabilities, and while they technically aren't, they basically are (see Measure Theory). For example, in Bayes':

$$Post(\beta|data) = \frac{Lik(data|\beta)P(\beta)}{\int \int Lik(data|\beta)P(\beta)d\beta}$$

where we are effectively treating likelihood as a probability.

$$L(\mathbf{y}) = \prod_{i=1}^n f(y_i) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2\sigma^2}(y_i - x_i^\top \beta)^2} = \left(\frac{1}{\sqrt{2\pi\sigma^2}}\right)^{n/2} e^{-\frac{1}{2\sigma^2} \sum_{i=1}^n (y_i - x_i^\top \beta)^2}$$

$$\sum_{i=1}^n (y_i - x_i^\top \beta)^2 = (\mathbf{y} - \mathbf{X}\beta)^\top (\mathbf{y} - \mathbf{X}\beta) \text{ This is then a dot-product of the two...}$$

$$\Rightarrow L(\mathbf{y}) = (2\pi)^{-n/2} (\sigma^2)^{-n/2} e^{-\frac{1}{2\sigma^2} (\mathbf{y} - \mathbf{X}\beta)^\top (\mathbf{y} - \mathbf{X}\beta)}$$

2 MLE of β

$$\underset{nx1}{\mathbf{y}} = \underset{nxp}{\mathbf{X}} \underset{px1}{\beta} + \underset{nx1}{\epsilon}, \epsilon_i \stackrel{\text{iid}}{\sim} N(0, \sigma^2), \mathbf{y} \sim N(\mathbf{X}\beta, \sigma^2 I_n)$$

$L(\mathbf{y}|\beta, \sigma^2)$ we just switch the conditioning around for the likelihood of the parameters:

$$L(\beta, \sigma^2|\mathbf{y}) = (2\pi)^{-n/2} (\sigma^2)^{-n/2} e^{-\frac{1}{2\sigma^2} (\mathbf{y} - \mathbf{X}\beta)^\top (\mathbf{y} - \mathbf{X}\beta)}$$

To find the MLE for β : $\frac{\partial}{\partial \beta} L(\beta, \sigma^2|\mathbf{y})$, set it equal to 0

$$l(\beta, \sigma^2|\mathbf{y}) = -\frac{n}{2} \ln(2\pi) - \frac{n}{2} \ln(\sigma^2) - \frac{1}{2\sigma^2} (\mathbf{y} - \mathbf{X}\beta)^\top (\mathbf{y} - \mathbf{X}\beta)$$

When we take the derivative we will get p items because there are p elements in β .

$$\frac{\partial}{\partial \beta} l(\beta, \sigma^2|\mathbf{y}) = -\frac{1}{2\sigma^2} \frac{\partial}{\partial \beta} (\mathbf{y} - \mathbf{X}\beta)^\top (\mathbf{y} - \mathbf{X}\beta)$$

$$(\mathbf{y} - \mathbf{X}\beta)^\top (\mathbf{y} - \mathbf{X}\beta) = \mathbf{y}^\top \mathbf{y} - \beta^\top \mathbf{X}^\top \mathbf{y} - \mathbf{y}^\top \mathbf{X} \beta + \beta^\top \mathbf{X}^\top \mathbf{X} \beta = \mathbf{y}^\top \mathbf{y} - 2\beta^\top \mathbf{X}^\top \mathbf{y} + \beta^\top \mathbf{X}^\top \mathbf{X} \beta$$

$$\frac{\partial}{\partial \beta} \mathbf{y}^\top \mathbf{y} - 2\beta^\top \mathbf{X}^\top \mathbf{y} + \beta^\top \mathbf{X}^\top \mathbf{X} \beta = -2\mathbf{X}^\top \mathbf{y} + 2\mathbf{X}^\top \mathbf{X} \beta$$

$$\text{Set it equal to 0: } -2X^\top \underset{\sim}{y} + 2X^\top X \underset{\sim}{\hat{\beta}} = 0$$

$$\implies X^\top X \underset{\sim}{\hat{\beta}} = X^\top \underset{\sim}{y}$$

$$\implies (X^\top X)^{-1} X^\top X \underset{\sim}{\hat{\beta}} = (X^\top X)^{-1} X^\top \underset{\sim}{y}$$

$$\implies \underset{\sim}{\hat{\beta}} = (X^\top X)^{-1} X^\top \underset{\sim}{y}$$

$$\mathbb{E}(\underset{\sim}{\hat{\beta}}) = \mathbb{E}[(X^\top X)^{-1} X^\top \underset{\sim}{y}] = (X^\top X)^{-1} X^\top X \underset{\sim}{\beta} = \underset{\sim}{\beta}$$

$$V(A \underset{\sim}{y}) = AV(\underset{\sim}{y})A^\top$$

$$Var(\underset{\sim}{\hat{\beta}}) = Var[(X^\top X)^{-1} X^\top \underset{\sim}{y}]$$

$$\underset{pxn}{A} = (\underset{pxp}{X^\top X})^{-1} \underset{pxn}{X^\top}$$

$$\underset{nxp}{A}^\top = X(X^\top X)^{-1}$$

$$Var(\underset{\sim}{\hat{\beta}}) = (X^\top X)^{-1} X^\top Var(\underset{\sim}{y}) X (X^\top X)^{-1} = (X^\top X)^{-1} X^\top \sigma^2 I_n X (X^\top X)^{-1} = \sigma^2 (X^\top X)^{-1}$$

$$\underset{px1}{\underset{\sim}{\hat{\beta}}} \sim N(\underset{px1}{\underset{\sim}{\beta}}, \underset{pxp}{\sigma^2 (X^\top X)^{-1}})$$

3 Multi-variate Normal

$$L = (2\pi)^{-n/2} |\Sigma|^{-1/2} e^{(-\frac{1}{2}(y-x\beta)^\top \Sigma^{-1} (y-x\beta))}$$

$$\Sigma = \sigma^2 I_n$$

$$L = (2\pi)^{-n/2} (\sigma^2)^{-n/2} e^{(-\frac{1}{2\sigma^2} (y-x\beta)^\top (y-x\beta))}$$

$$l = -\frac{n}{2} \ln(2\pi) - \frac{1}{2} \ln(|\Sigma|) - \frac{1}{2} (y-x\beta)^\top \Sigma^{-1} (y-x\beta)$$

$$\frac{\partial}{\partial \beta} (y-x\beta)^\top \Sigma^{-1} (y-x\beta)$$

$$\text{HW1 shows the derivation of the MLE for } \beta \text{ in the Multi-variate normal: } \hat{\beta} = (x^\top \Sigma^{-1} x)^\top x^\top \Sigma^{-1} y$$

$$\mathbb{E}[\hat{\beta}] = \mathbb{E}_y[(x^\top \Sigma^{-1} x)^\top x^\top \Sigma^{-1} y] = (x^\top \Sigma^{-1} x)^\top x^\top \Sigma^{-1} \mathbb{E}[y] = (x^\top \Sigma^{-1} x)^\top x^\top \Sigma^{-1} x \beta = \beta$$

We also get an unbiased by ignoring the covariance:

$$\mathbb{E}[(x^\top x)^{-1} x^\top y] = \beta$$

$$\text{The variance for this (excluding covariance) is: } Var(\hat{\beta}) = \hat{\sigma}^2 (x^\top x)^{-1}$$

Now, finding the Variance for the actual estimate (including covariance):

(remember that $Var(y) = \Sigma$)

$$Var[\hat{\beta}] = Var((x^\top \Sigma^{-1} x)^{-1} x^\top \Sigma^{-1} y) = (x^\top \Sigma^{-1} x)^{-1} x^\top \Sigma^{-1} Var(y) \Sigma^{-1} x (x^\top \Sigma^{-1} x)^{-1} = (x^\top \Sigma^{-1} x)^{-1}$$

4 Cell Means Stuff

Going to back to OG normals:

$$\underset{\sim}{\mathbf{y}} \sim N(\underset{\sim}{\boldsymbol{\mu}}, \sigma^2 I), \text{ which we can just write as } N(x\underset{\sim}{\boldsymbol{\mu}}, \sigma^2 I_n)$$

This would be x is just a column of 1's: $(x^\top x)^{-1} x^\top \underset{\sim}{\mathbf{y}} = \frac{1}{n} \sum y_i = \bar{y}$

$$\text{Now let } \underset{\sim}{\boldsymbol{\mu}} = \begin{pmatrix} \mu_1 \\ \mu_2 \end{pmatrix}$$

$$\text{The y is now a combined vector of data from each: } \begin{pmatrix} y_{11} \\ y_{12} \\ \vdots \\ y_{1n_1} \\ y_{21} \\ y_{22} \\ \vdots \\ y_{2n_2} \end{pmatrix}$$

$$\text{So now x is a 2 column 'identity' matrix showing which mean each datapoint came from: } \begin{pmatrix} 1 & 0 \\ 1 & 0 \\ \vdots & \vdots \\ 1 & 0 \\ 0 & 1 \\ 0 & 1 \\ \vdots & \vdots \\ 0 & 1 \end{pmatrix}$$

$$(x^\top x) = \begin{pmatrix} n_1 & 0 \\ 0 & n_2 \end{pmatrix}$$

$$(x^\top x) = \begin{pmatrix} \frac{1}{n_1} & 0 \\ 0 & \frac{1}{n_2} \end{pmatrix}$$

$$x^\top \underset{\sim}{\mathbf{y}} = \begin{pmatrix} \sum_{i=1}^{n_1} y_{1i} \\ \sum_{i=1}^{n_2} y_{2i} \end{pmatrix}$$

t-test:

$$\frac{\text{est} - \text{null value}}{s.e.(\text{est})} = \frac{\bar{y} - \text{null}}{\sqrt{\hat{\sigma}^2 (x^\top x)^{-1}}} = \frac{(y - \hat{y})^\top (y - \hat{y})}{n-1} = \frac{\bar{y} - \text{null}}{\sqrt{\frac{\hat{\sigma}^2}{n}}} = \frac{\bar{y} - \text{null}}{\frac{\hat{\sigma}}{\sqrt{n}}}$$

Two Sample Independent t-test: null is $\mu_1 = \mu_2$:

$$\text{Let our y's be: } \begin{pmatrix} \bar{y}_1 \\ \bar{y}_2 \end{pmatrix}, \text{ and c be } \begin{pmatrix} 1 \\ -1 \end{pmatrix}, \text{ then } \underset{\sim}{\mathbf{c}} \underset{\sim}{\mathbf{y}} = \bar{y}_1 - \bar{y}_2$$

$$Var(c\hat{\beta}) = cVar(\hat{\beta})c^\top = \begin{pmatrix} 1 & -1 \end{pmatrix} \hat{\sigma}^2 (x^\top x)^{-1} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} 1 & -1 \end{pmatrix} \hat{\sigma}^2 \begin{pmatrix} \frac{1}{n_1} & 0 \\ 0 & \frac{1}{n_2} \end{pmatrix} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \hat{\sigma}^2 \left(\frac{1}{n_1} + \frac{1}{n_2} \right)$$

$$\frac{\text{est} - \text{null}}{\text{s.e}(\text{est})} = \frac{c\hat{\beta}}{\text{se}(c\hat{\beta})} = \frac{\bar{y}_1 - \bar{y}_2}{\sqrt{\hat{\sigma}^2(\frac{1}{n_1} + \frac{1}{n_2})}}$$

$$F_{1,n_1+n_2-2} = \frac{(\bar{y}_1 - \bar{y}_2)}{\hat{\sigma}^2(\frac{1}{n_1} + \frac{1}{n_2})}$$

Now we use more x's:

$$\underset{\sim}{\mathbf{y}} \sim N(x\beta, \sigma^2 I_n)$$

$$\underset{\sim}{\boldsymbol{\mu}} = \begin{pmatrix} \mu_1 \\ \mu_2 \\ \mu_3 \end{pmatrix}$$

$$x = \begin{pmatrix} 1 & 0 & 0 \\ \vdots & \vdots & \vdots \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ \vdots & \vdots & \vdots \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ \vdots & \vdots & \vdots \\ 0 & 0 & 1 \end{pmatrix}$$

$$\hat{\mu} = (x^\top x)^{-1} x^\top y = \begin{pmatrix} \frac{1}{n_1} \sum_{i=1}^{n_1} y_{1i} \\ \frac{1}{n_2} \sum_{i=1}^{n_2} y_{2i} \\ \frac{1}{n_3} \sum_{i=1}^{n_3} y_{3i} \end{pmatrix}$$

$$c = \begin{pmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \end{pmatrix}$$

$$c\underset{\sim}{\boldsymbol{\mu}} = \begin{pmatrix} \mu_1 - \mu_2 \\ \mu_2 - \mu_3 \end{pmatrix}$$

$$F_{\text{rank of } c, n-p} = (c\hat{\beta})^\top (c\text{Var}(\hat{\beta})c^\top)^{-1} (c\hat{\beta}) \frac{1}{\text{rank of } c}$$

$$\text{null: } \mu_1 = \mu_2 = \mu_3$$

5

$$\text{Likelihood: } (2\pi)^{-n/2} |\Sigma|^{-1/2} e^{-\frac{1}{2}(y-x\beta)^\top \Sigma^{-1} (y-x\beta)}$$

$$\text{Max Likelihood for } \hat{\beta} = (X^\top \Sigma^{-1} X)^{-1} X^\top \Sigma^{-1} \underset{\sim}{\mathbf{y}}$$

$$V(\hat{\beta}) = (X^\top \Sigma^{-1} X)^{-1}$$

For the Ingot bond data, think of σ_I^2 as the variance between ingots, and σ_e^2 as the variance within a single ingot.

$$\begin{pmatrix} y_1 \\ y_2 \end{pmatrix}: V(\underset{\sim}{\mathbf{y}}) = \Sigma = \begin{pmatrix} \sigma_e^2 + \sigma_I^2 & \sigma_I^2 \\ \sigma_I^2 & \sigma_e^2 + \sigma_I^2 \end{pmatrix}$$

$$\begin{pmatrix} y_{11} \\ y_{12} \\ y_{21} \\ y_{22} \end{pmatrix} : V(\underset{\sim}{\mathbf{y}}) = \Sigma = \begin{pmatrix} \sigma_e^2 + \sigma_I^2 & \sigma_I^2 & 0 & 0 \\ \sigma_I^2 & \sigma_e^2 + \sigma_I^2 & 0 & 0 \\ 0 & 0 & \sigma_e^2 + \sigma_I^2 & \sigma_I^2 \\ 0 & 0 & \sigma_I^2 & \sigma_e^2 + \sigma_I^2 \end{pmatrix}$$

$$|\Sigma| = (\sigma_e^2 + \sigma_I^2) - (\sigma_I^2)^2$$

$$\text{Remember, if } A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, A^{-1} = \frac{1}{|A|} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

$$\Sigma^{-1} = \begin{pmatrix} \frac{\sigma_e^2 + \sigma_I^2}{(\sigma_e^2 + \sigma_I^2) - (\sigma_I^2)^2} & \frac{-\sigma_I^2}{(\sigma_e^2 + \sigma_I^2) - (\sigma_I^2)^2} \\ \frac{-\sigma_I^2}{(\sigma_e^2 + \sigma_I^2) - (\sigma_I^2)^2} & \frac{\sigma_e^2 + \sigma_I^2}{(\sigma_e^2 + \sigma_I^2) - (\sigma_I^2)^2} \end{pmatrix}$$

$$\text{If } X = \begin{pmatrix} x_1^\top \\ x_2^\top \\ \vdots \\ x_n^\top \end{pmatrix} \text{ then } (y - x\beta) = \begin{pmatrix} y_1 - x_1^\top \beta \\ y_2 - x_2^\top \beta \end{pmatrix}$$

6

Fixed Factor: all the levels of that factor are known. Not trying to make inference for anything outside that range/given levels.

Random Factor: we want to make inference to other things, generalize to the population, etc.

Generalized Least Squares (GLS)

$$\hat{\beta} = (X^\top \hat{\Sigma}^{-1} X)^{-1} X^\top \hat{\Sigma}^{-1} \underset{\sim}{\mathbf{y}}$$

$y = x\underset{\sim}{\beta} + z\underset{\sim}{\mathbf{u}} + \underset{\sim}{\epsilon}, \underset{\sim}{\epsilon} \sim N(\underset{\sim}{\mathbf{0}}, R), \underset{\sim}{\mathbf{u}} \sim N(\underset{\sim}{\mathbf{0}}, G), \epsilon, u$ are independent, where x is the fixed effect and z is the random effect. Hence the term mixed models

Fixed effects coefficients tend to be greek letters, and random coefficients tend to be latin letters.

$$y : 21x1, x : 21x3, z : 21x7$$

In the Ingot example, $G = \sigma_I^2 I_I$

3 is the number of β s, 7 is the number of u s.

$$\mathbb{E}[\underset{\sim}{\mathbf{y}}] = \mathbb{E}[x\underset{\sim}{\beta} + z\underset{\sim}{\mathbf{u}} + \underset{\sim}{\epsilon}] = x\beta$$

$$Var(y) = V(x\underset{\sim}{\beta} + z\underset{\sim}{\mathbf{u}} + \underset{\sim}{\epsilon}) = xV(\beta)x^\top + zVar(u)z^\top + Var(\epsilon) = zGz^\top + R$$

$$\begin{pmatrix} y_1 = m_1 + u_1 + \epsilon_1 \\ y_2 = m_2 + u_1 + \epsilon_2 \\ y_3 = m_3 + u_1 + \epsilon_3 \\ y_4 = m_1 + u_2 + \epsilon_4 \\ \vdots \\ y_{21} = m_3 + u_7 + \epsilon_{21} \end{pmatrix}$$

$$z = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ & & & \vdots & & & \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

HW: write out the X matrix, Z matrix, G matrix, R matrix, zGz' + R matrix? $G = \sigma_I^2 I_1 R = \sigma_e^2 I_{21} X = \text{Cell Means}$